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Interpretation of Surface EMGs in Children with Cerebral Palsy: An Initial Study Using a Fuzzy Expert System

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ABSTRACT: Surface EMG detected simultaneously at different muscles has become an important tool for analysing the gait of children with cerebral palsy (CP), as it offers essential information about muscular coordination. However, the interpretation of surface EMG is a difficult task that assumes extensive knowledge and experience. As such, this noninvasive procedure is not frequently used in the general clinical routine. An Artificial Intelligence (AI) system for interpreting surface EMG signals and the resulting muscular coordination patterns could overcome these limitations. To support such interpretation, an expert system based on fuzzy inference methodology was developed. The knowledge-base of the system implemented 15 rules, from which the fuzzy inference methodology performs a prediction of the effectiveness of the muscular coordination during gait. Our aim was to assess the feasibility and value of such an expert system in clinical applications. Surface EMG signals were recorded from the tibialis anterior, soleus muscle, and gastrocnemius muscles of children with CP to assess muscular coordination patterns of ankle movement during gait. Nineteen children underwent 114 surface EMG measurements. Simultaneously, the gait cycles of each patient were determined using foot switches and videotapes. From the EMG signals, the effectiveness of the ankle movement was predicted by the expert system, and predictions were classified using a three-point ordinal scale. In 91 cases (80%), the clinical findings matched the predictions of the expert system. In 23 cases (20%) the predictions of the expert system differed from the clinical findings with 12 cases revealing worse and 11 cases revealing better results in comparison to the clinical findings. As this study is a first attempt to verify the feasibility and correctness of this expert system, the results are promising. Further study is required to assess the correlation with the kinematic data and to include the whole leg. © 2006 Orthopaedic Research Society. Published by Wiley Periodicals, Inc. J Orthop Res 24:438–447, 2006

Keywords: surface EMG; gait analysis; cerebral palsy; expert system; fuzzy logic

INTRODUCTION

Gait analysis is a valuable tool in the treatment of cerebral palsy (CP), providing a precise assessment of gait abnormalities. Surface EMG is an essential part of gait analysis, providing the physician with objective information about muscle function and insight into muscle recruitment by the central nervous system.[1–4] A major finding in CP is abnormal muscle tone and imbalance between muscle agonists and antagonists. Hence, surface EMG provides the opportunity to assess the activity of different muscles during the complex variations of coordination.[5] In recent years, surface EMG investigations have become a tool in planning treatment for children with CP

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and for postoperative assessment of the effectiveness of surgical procedures.

Nevertheless, the results obtained with surface EMG must be interpreted by the clinician. As information regarding muscular coordination is based on the activation pattern of different muscles, several EMG signals must be recorded synchronously, and the resulting signals cannot be independently interpreted. Consequently, the signal interpretation process is complex, and depends essentially on the clinician's expertise. Most interpretation is performed visually and qualitatively; no system is available to support the clinician in the process of interpretation and decision making.

Expert systems are designed to support the process of decision making. Recently, they have been implemented successfully in the medical field to support interpretation of biophysical signals of different origin.[6–9] An expert system is defined as an artificial system that facilitates solving problems in a given field or application by drawing inference from a knowledge base developed from human expertise. Therefore, an expert system consists of a knowledge base, which contains the human expertise, and an inference engine, which processes the input variables based on the knowledge base and merges them to one or more meaningful output variables (Fig. 1).

However, human expertise and the decision-making process are seldom precisely specified. This becomes a serious problem when muscular coordination must be interpreted on the basis of surface EMG signals. The surface EMG signals themselves and their interpretation contain some degree of uncertainty that makes the definition of precise rules for decision making impossible. The fuzzy logic or fuzzy sets theory is tolerant of imprecise data and, therefore, well suited to build an expert system for interpreting muscular coordination based on surface EMG.[10,11] Compared to other methodologies, a fuzzy inference system provides several advantages. Fuzzy logic makes the management of uncertain, complex, logical connection with linguistic instruments possible.[12–15] Additionally, the basis of fuzzy logic is a natural language that is also the basis of human communication. The use of linguistic variables facilitates knowledge acquisition, especially from nontechnical experts, and the integration of different types of knowledge from various experts.

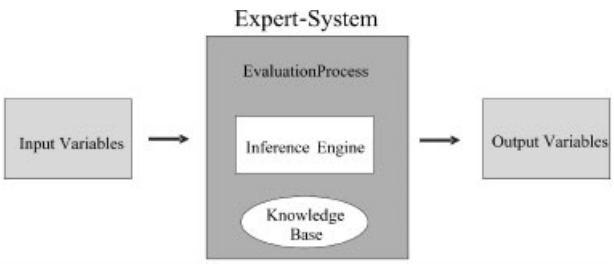

Figure 1. Components of an expert system. The input variables are processed by the inference engine supported by a knowledge base. The output variables represent an interpretation of the input variables.

An expert system has been created that has been, in a first attempt, exemplified by the muscular coordination pattern of the shank muscles during gait. With the help of this expert system, we tried to obtain one curve instead of three EMG curves (from three different muscles), which should provide information about the effectiveness of muscular coordination with respect to ankle movement. The aim of the presented study was to verify the feasibility of this new expert system and to determine if a correlation exists between clinical findings in patients with CP and the results predicted by the expert system.

PATIENTS AND METHODS

Data Acquisition

The investigations were performed on 19 children with cerebral palsy: 6 girls (32%) and 13 boys (68%). Ages ranged from 3 to 12 years (mean = 7.8 years). The majority of patients were diplegic ($n = 15$, 79%); four children (21%) were hemiplegic. Children who were unable to walk with minor support (a helping hand) and those who were uncooperative were excluded. When performing the gait cycles no orthoses, canes, or crutches were used. Informed consent was obtained from the parents. Three of the children had undergone previous surgery on their legs and hips. All had experienced various forms of physiotherapy. EMG examinations were performed before local application of botolinum toxin or other therapies were considered.

Bipolar surface EMG was recorded synchronously from the tibialis anterior, gastrocnemius, and soleus muscles. Adhesive Ag/AgCl electrodes with a diameter of 10 mm and an interelectrode distance of 20 mm were used. The electrodes were localized according to the SENIAM recommendations.[16,17] While the electrodes for the gastrocnemius were placed on the most prominent bulge of the muscle, those for the soleus were positioned 2/3 of the distance along an imaginary line joining the medial condyle of the femur and the medial malleolus. The electrodes of the tibialis anterior muscle were positioned 1/3 of the distance along an imaginary line joining the tip of the fibula and the tip of the medial

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malleolus. The signals were amplified (factor 1000), band-pass filtered (10–250 Hz), and sampled at a frequency of 1000 Hz. Foot switches were simultaneously used to define the gait cycle. The raw EMG data were visually assessed for artefacts and noise. Signals that showed clear artefacts, especially movement artefacts, or noise inference of the base line were excluded. The EMG signals were subsequently rectified and smoothed by a 30-Hz low-pass filter.Different EMG sweeps were synchronized to the gait cycle and averaged for each muscle. Additionally, the gait of each patient was videotaped. Six consecutive EMG examinations, three on each side, were performed. When artefacts occurred or when the children stopped during a measurement, additional examinations were performed. Every EMG examination was obtained over a walking distance of about 10 m. Afterwards, each EMG examination was evaluated by the expert system. Six predictions of the effectiveness of ankle joint movement per patient, three of each side, were separately obtained.

Design of the Expert System

The expert system consisted of an inference engine, which processed input variables using a knowledge base of expertise. In doing so, more meaningful output variables were generated (Fig. 1). The fuzzy inference used to support the interpretation of muscular coordination is called Mamdani-style inference.[18] As input variables, the processed surface EMGs of the tibialis anterior, gastrocnemius, and soleus muscles were used.

The knowledge base was built using knowledge acquired from three different orthopedic surgeons who were well trained in the interpretation of muscular coordination during gait. Additionally, information from the literature was included.[19,20] On the basis of knowledge acquisition and information from the literature, a total of 15 rules were developed (Table 1), all having a form similar to:

If "x" is high and "y" is low then the "z" is high

where x and y are input variables, and z is the output variable of the expert system. The input variables of the fuzzy expert system used to support the interpretation of muscular coordination of the ankle joint during gait were called *TibAnt, Gastroc*, and *Soleus*, and corresponded to the averaged envelopes of the surface EMGs of the muscles (Fig. 2). In the case of muscular activation, the condition "high" was equal to "the muscle is active," and the condition "low" corresponded to the negation of "high," which means "inactive." Movements arising as a result of differences in muscle strength were

Table 1. Fifteen Rules used to Construct the Knowledge Base of the Expert System

No.	Rule	Weight
1	If *TibAnt* is active and *Gastroc* is not active and *Soleus* is not active, then *Effectiveness of Movement = dorsiflexion* very high	0.8
2	If *TibAnt* is not active and *Gastroc* is active and *Soleus* is active, then *Effectiveness of Movement = plantarflexion* very high	0.8
3	If *TibAnt* is not active and *Soleus* is active, then *Effectiveness of Movement = plantarfexion* high	0.5
4	If *TibAnt* is not active and *Gastroc* is active, then *Effectiveness of Movement = plantarflexion* high	0.5
5	If *TibAnt* is not active and *Gastroc* is not active and *Soleus* is not active, then *Effectiveness of Movement = plantarflexion* low	0.2
6	If *TibAnt* is active and *Soleus* is active, then *Effectiveness of Movement = plantarflexion* low	0.2
7	If *TibAnt* is active and *Gastroc* is active, then *Effectiveness of Movement = plantarflexion* low	0.2
8	If *TibAnt* is active and *Gastroc* is active and *Soleus* is active, then *Effectiveness of Movement = plantarflexion* very low	0.4
9	If *TibAnt* is active and *Soleus* is active, then *Effectiveness of Movement = dorsiflexion* low	0.2
10	If *TibAnt* is active and *Gastroc* is active, then *Effectiveness of Movement = dorsiflexion* low	0.2
11	If *TibAnt* is active and *Gastroc* is active and *Soleus* is active, then *Effectiveness of Movement = dorsiflexion* very low	0.2
12	If *TibAnt* is active between 0–15% of the gait cycle, then *Effectiveness of Movement = dorsiflexion* very high	0.8
13	If *TibAnt* is active between 55–100% of the gait cycle, then *Effectiveness of Movement = dorsiflexion* very high	0.4
14	If *Gastroc* is active between 20–55% of the gait cycle, then *Effectiveness of Movement = plantarflexion* high	0.7
15	If *Soleus* is active between 15–55% of the gait cycle, then *Effectiveness of Movement = plantarflexion* high	0.5

A prediction of the effectiveness of ankle joint movement is performed by the fuzzy inference based on these rules. Dorsiflexion and plantarflexion must be handled separately.

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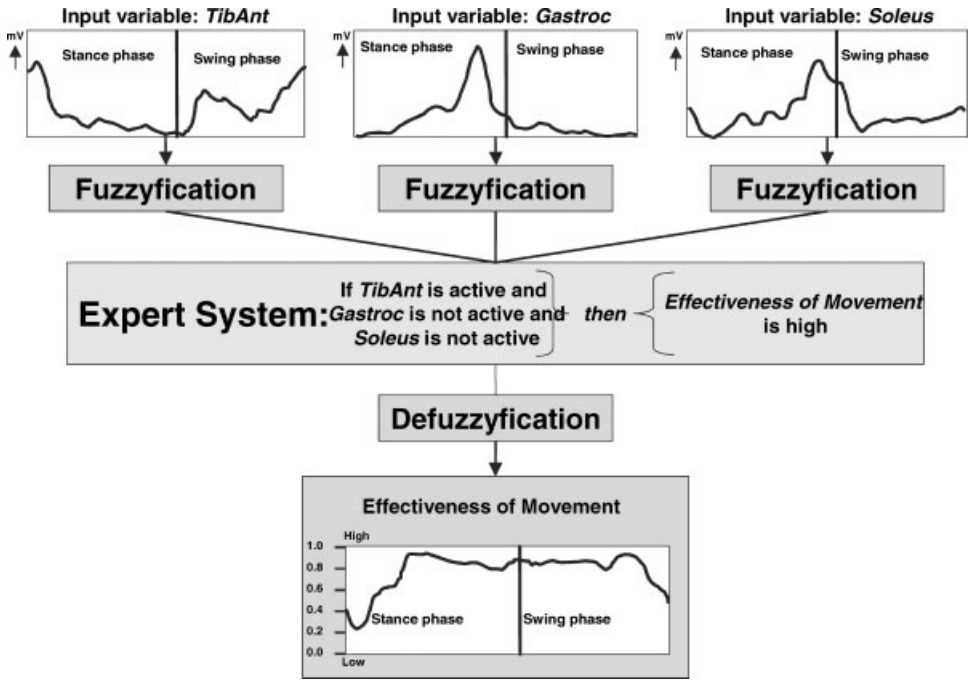

Figure 2. Processing of the input variables *TibAnt*, *Gastroc*, and *Soleus* by the expert system. One rule has been included as an example. One characteristic course of the output variable *Effectiveness of Movement* is presented.

considered in the uncertainty of the fuzzy inference. The output variable was the prediction of whether ankle dorsiflexion and plantarflexion would take place. Those cases in which a coactivation was needed to stabilize the joint were incorporated in the rules.

As usual in fuzzy logic, membership functions defined whether an input variable belonged to each of the appropriate fuzzy sets. In the case of interpreting muscular coordination, only one fuzzy set "active" for each muscle had to be developed referring to the degree of activity of a certain muscle. The second required fuzzy set "inactive" was the negation of the first and could be built from the first one by a fuzzy NONoperation. The step to determine the degree by which an input variable belonged to a fuzzy set is called fuzzification (Fig. 2). By consulting different experts conversant with the interpretation of surface EMG signals, the membership functions used for fuzzification were developed. The membership functions of all three muscles were constructed in a similar way. A muscle was defined as active if the EMG value exceeded a given starting threshold. With increasing EMG value, the degree of activity increased proportionally until a second threshold was reached. For all EMG values higher than the second threshold, the membership function became one, meaning that the muscle was completely activated (Fig. 3). In the case of the tibialis anterior muscle, the starting threshold was chosen as 10% and the final threshold as 70% of the maximum EMG value occurring during the gait cycle. Starting activation of the gastrocnemius and soleus muscles was defined at 20% and maximum activation at 80% of the maximal measured EMG value of the specific muscle.

The different starting and final thresholds of the three muscles reflect the fact that the surface EMG signals recorded from various muscles have different signal-to-noise ratios that can differ between patients. Soleus and gastrocnemius were placed more distantly from the recording electrodes in comparison to the tibialis anterior muscle. Consequently, the EMG signal of the tibialis anterior muscle was less affected by individual factors (e.g., body mass). This higher interindividual stability of tibialis anterior EMG was considered by shifting the threshold toward lower values. By means of a fuzzy AND operation from the three membership functions and their negations, one antecedent value was obtained. Based on this antecedent value, two implication variables called *plantarflexion* and *dorsiflexion* (Table 1 and Fig. 3) were assessed. Whether one of them was high or low was also defined in membership functions (Fig. 3).

The membership functions of the variables were optimized with the help of 50 healthy gait cycles, assuming that in healthy gait the pattern of ankle movement was most effective. The so-called implication process involves the realization of the consequences that arise from the combination of the input variables on the basis of the knowledge base. Both the generation of the antecedent value from the EMG input variables and the implication process were separately performed for

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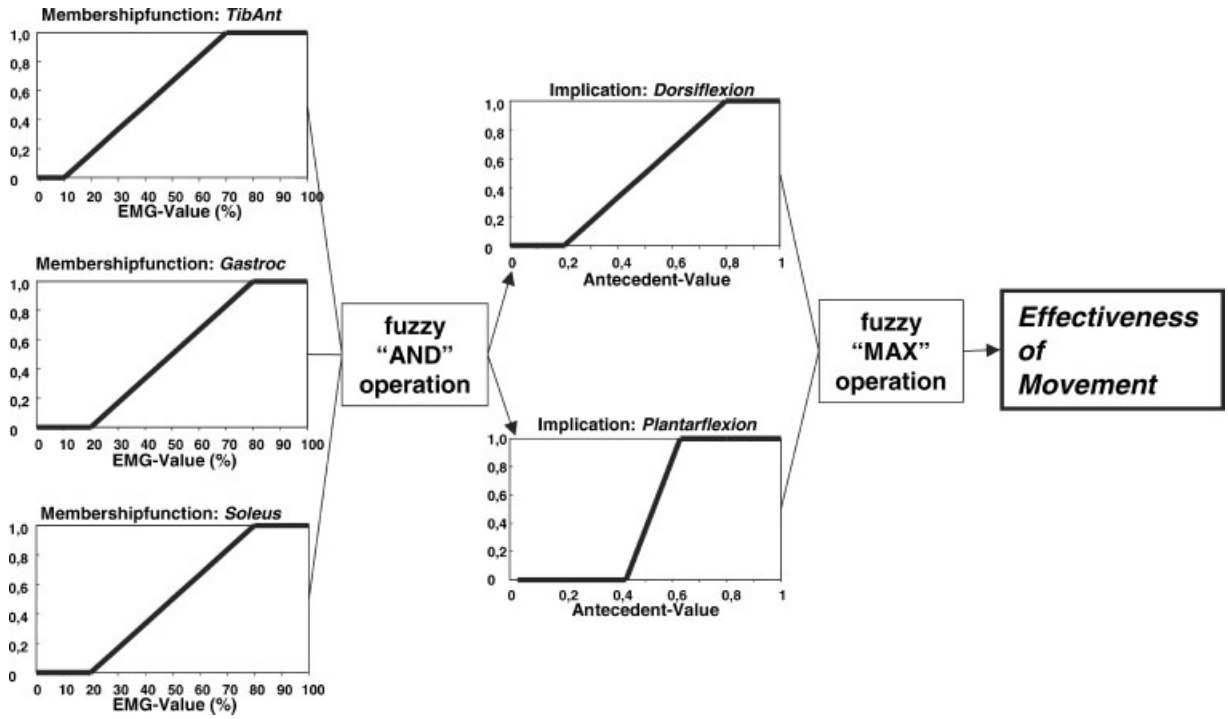

Figure 3. Membership functions of the input variables *TibAnt*, *Gastroc,* and *Soleus* and the implementation functions for the variables *dorsiflexion* and *plantarflexion*. The depicted processes are separately performed for each rule. Afterward, an agglomeration process is performed including all rules in one output variable.

every rule of the knowledge base. The different rules were weighted with respect to their contribution to the outcome variable. The slopes and the thresholds of the membership functions of *dorsiflexion* and *plantarflexion* as well as the different weighting factors were optimized on healthy gait cycles.

The combination of all 15 rules was performed by an aggregation process. Each rule led to one fuzzy set describing the functional output with respect to *dorsiflexion* and *plantarflexion*. The aggregation process was performed by a MAX operation, where for each antecedent value the maximum value of the implications of all rules was used (Fig. 4). With a fuzzy MAX operation, the two implication variables regarding ankle *dorsiflexion* and *plantarflexion* could be merged into one output variable (Fig. 3). This output variable was initially also linguistic information like "the effectiveness of the movement is high" (Fig. 2). Consequently, it needs defuzzification to be transformed back into crisp values ranging from 0 (poor effectiveness) to 1 (high effectiveness). Thus, the centroid calculation, which returns the center of the area under the curve resulting from the aggregation process, was chosen to effect defuzzification. The resulting output variable, *Effectiveness of Movement*, marks the end of the fuzzy process and was calculated separately for each percent interval of the gait cycle. The output variable indicates to what extent and at what periods of the gait cycle the muscular coordination pattern of the shank muscles yielded an effective movement of the ankle joint during gait.

Comparison of Clinical Findings and the Prediction of the Expert System

Because *Effectiveness of Movement* showed different characteristics for different patients, a scoring system was developed that allowed this variable to be compared with the clinical findings. The assessment of the curves given by the expert system revealed three typical patterns (Fig. 5). Therefore, a scoring system was established using a three-point ordinal scale. The three EMG stages were grossly categorized by observation of the program output. In stage I, the curve representing *Effectiveness of Movement* remained between 0.7 and 0.8, and did not drop below 0.6 (Fig. 5). In stage II, one major drop occurred well below 0.6 with a sharp rise immediately after the drop, while in stage III, the curve remained on an average below 0.6 (Fig. 5).

Step length, stance phase, range of motion (ROM) of ankle and knee joint while walking, fixed contractures of the knee or ankle, use of the upper limbs for stabilization, and ability to reach the ground with the heels during stance phase were considered in defining the clinical gait of the patients. On this basis, clinical

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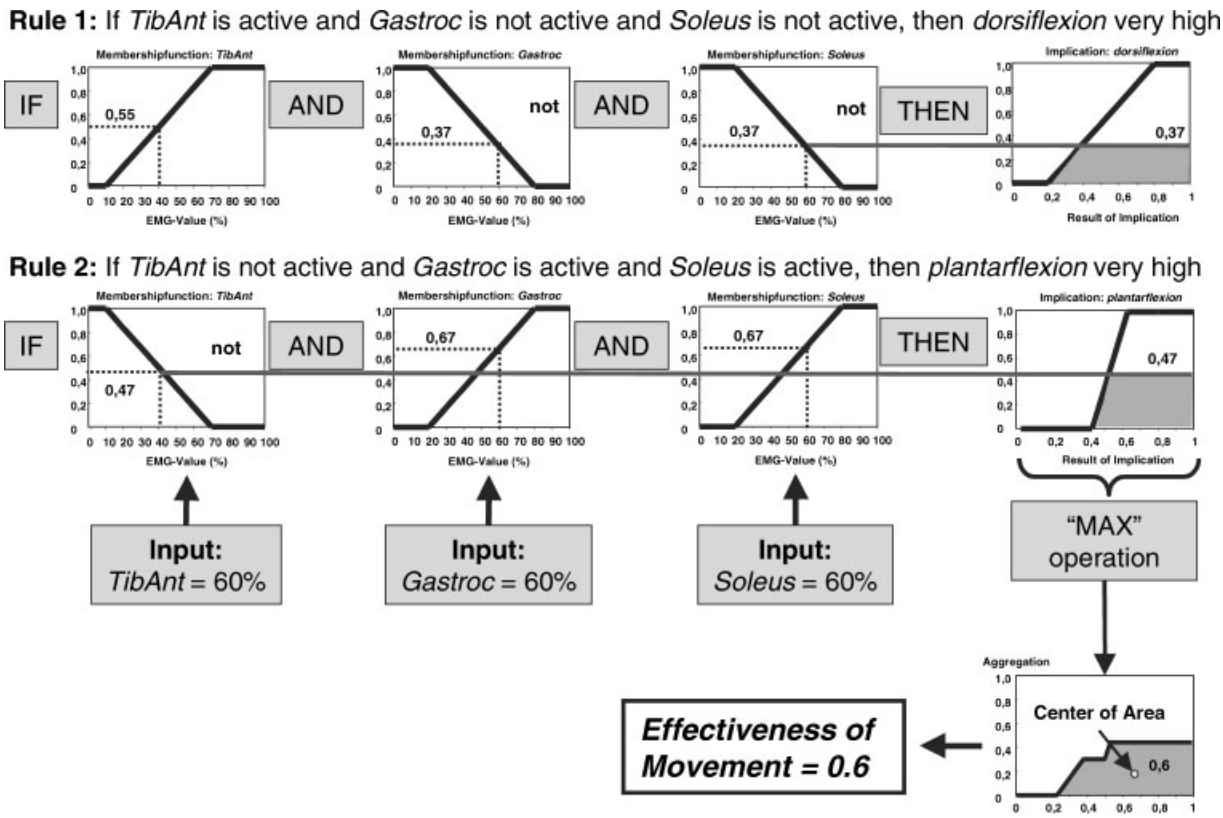

Figure 4. Illustration of the agglomeration process for Rule 1 and Rule 2

findings were divided into three categories. Clinical stage I included patients who could walk safely without external support from orthoses or a helping hand. The walking patterns were fairly smooth. When walking at normal speed, the heels reached the ground. The ROMs were more or less normal with no major contractures of the hip, knee, or ankle. Clinical stage II comprised patients whose gait showed at least intermittent contact of the heels with the ground. Standing was possible without support. The ROMs were compromised only to minor degrees. In Clinical stage III, patients required intermittent support while walking. The heels did not reach the ground and the ROMs were severely limited. They were unable to walk backwards or take steps of normal height. Many of them showed a crouched pattern when walking. If a child did not meet all criteria for stage I, he/she was assigned to stage II; the same strategy applied to stage II.

The assignment of each child to a clinical stage was performed without knowledge of the results of the expert system and vice versa. Each side of each patient was assessed and graded separately as side-dependent differences exist in all patients. Additionally, some diplegic children showed side-dependent differences in the spasticity of their legs.

RESULTS

On the basis of the clinical findings and with respect to each leg, we found a clinical stage I

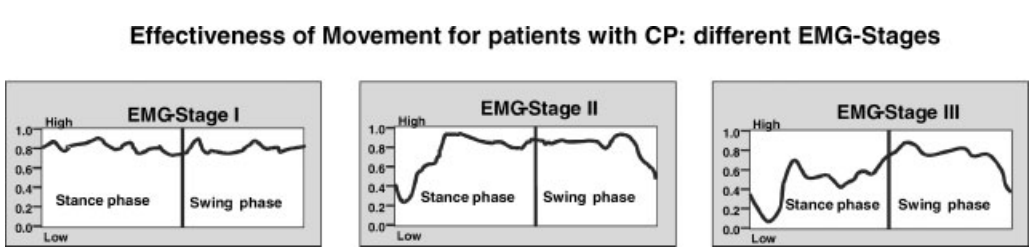

Figure 5. Output variable *Effectiveness of Movement* of the expert system for three patients of different stages. Typical progressions of the output variable are shown, which can be classified in three groups (EMG stage I–EMG stage III).

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8 times, a stage II 11 times, and a stage III 19 times (Table 2). With respect to the patients, two children were bilaterally classified as stage I, three patients as stage II, and eight patients as stage III. We found side-dependent differences in six children. The results from the expert system also showed predominantly stage III (Table 2). However, identical results were not observed for each leg in every child when considering the three trials. Different results were predicted on 10 occasions with the expert system.

In 91 of the 114 cases (80%), the predictions by the expert system matched the results of the clinical findings (Table 2, right-hand column). In

Table 2. Comparison of Clinical Findings with those Given by the Expert System

			Output of Fuzzy Expert System			
Patient	Side of the Leg	Clinical Categorization	EMG Stage I	EMG Stage II	EMG Stage III	Incorrect Findings
1	R	II	0	3	0	0
	L	I	3	0	0	0
2	R	II	0	3	0	0
	L	II	0	3	0	0
3	R	III	0	0	3	0
	L	III	0	0	3	0
4	R	III	0	3	0	3
	L	III	0	1	2	1
5	R	III	0	0	3	0
	L	II	0	2	1	1
6	R	II	0	0	3	3
	L	II	1	2	0	1
7	R	I	2	1	0	1
	L	II	0	3	0	0
8	R	II	1	2	0	1
	L	I	2	1	0	1
9	R	III	0	1	2	1
	L	III	0	0	3	0
10	R	II	0	3	0	0
	L	II	0	3	0	0
11	R	III	0	0	3	0
	L	III	0	0	3	0
12	R	II	0	3	0	0
	L	III	0	0	3	0
13	R	III	1	1	1	2
	L	III	0	1	2	1
14	R	I	3	0	0	0
	L	I	3	0	0	0
15	R	III	0	0	3	0
	L	III	0	0	3	0
16	R	I	0	3	0	3
	L	I	0	3	0	3
17	R	III	0	0	3	0
	L	III	0	0	3	0
18	R	III	0	0	3	0
	L	III	0	0	3	0
19	R	I	3	0	0	0
	L	III	0	1	2	1

The categorization of the clinical findings is presented in Roman numerals; the numbers for every EMG stage are given in Arabic. In addition, the number of discrepancies between clinical and EMG results are presented on the right-hand column.

$\Sigma = 23$ (20%).

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12 cases, the predictions of the expert system were "worse" when compared with the clinical findings, that is, the stage given by the expert system suggested a greater level of impairment than the medically assigned clinical stage; eight cases with a clinical stage I were an EMG stage II and four cases with clinical stage II were an EMG stage III by the expert system. By contrast, in 11 other cases the predictions of the expert system revealed "better" results than the clinical findings suggested; eight clinical stage III findings were an EMG stage II, while in one clinical stage III case an EMG stage I was predicted. Finally, two clinical stage II findings were rated as stage I by the expert system.

The processed EMG results matched the clinical stages in 10 patients. Conversely, in nine patients, at least one of the expert system predictions differed from the clinical classifications. However, a complete lack of correspondence occurred in only one patient, that is, all six EMG-based predictions differed from the clinical classification. With regard to the clinical stages, incorrect predictions of the expert system were obtained eight times for clinical stage I, six times for clinical stage II, and nine for stage III (Table 3).

DISCUSSION

Gait analysis including surface EMG in children with CP provides better assessment of clinical findings, improves preoperative decisions, and makes clinical findings more suitable for inter- and intraindividual comparisons.[21,22] Furthermore, gait analysis is important for an accurate and objective assessment of the outcome after surgical or other therapeutic procedures.[23] Especially in children with CP, information about muscular coordination patterns is important. Surface EMG can easily provide this information, although it is not frequently used in clinical routine. This may be attributed to the fact that surface EMG is often difficult to interpret, even by experienced physicians, motivating the introduction of an expert system to support interpretation of surface EMG data in the assessment of muscular coordination pattern.[11]

A knowledge-based system is desirable because it facilitates the evaluation of results independent from the knowledge, experience, and accuracy of a single interpreter. In the past, attempts have been made to compare computer-averaged EMG profiles with processed surface EMG data from healthy individuals.[24] This technique was intended to simplify the comparison with "normal" EMGs and to determine if a single EMG signal was normal. However, this system was fairly complex, and was unable to combine information from several EMG channels detected synchronously at different muscles. Consequently, it never achieved widespread use in the general clinical routine.

The present study was performed to assess the feasibility of a fuzzy expert system for the interpretation of EMG data from patients with CP and to ascertain the accuracy of the results. As a first step, the approach was restricted to the lower leg and the ankle joint. Such a restriction, from a clinical viewpoint, is somewhat artificial, as the clinical features of the shank and the ankle movement are influenced by other parts of the leg, including the knee and hip joints and the thigh. This influence might be a potential cause for some of the discrepancies observed between the predictions of the expert system and the clinical findings. Nevertheless, at this preliminary stage the limitation was helpful to assess whether this expert system might be a useful and sensible tool.

Another possible limitation was the clinical classification system, which was not validated and did not include the exact degree of the muscle tone. Therefore, we cannot rule out inaccuracy in this aspect of the study. Several systems are available for the classification of clinical findings in patients with CP including the Ashworth scale and the Gross Motor Function Measurement.[25–27] However, none of them has gained unrestricted acceptance, and as such, all were not suitable for our purposes. Another option would have been the comparison of the data generated by the expert system with the kinematics data. However, as the EMG data focus on the muscular coordination and not on the kinematic aspects, we abandoned this

Table 3. Correctness of EMG-based Predictions in Relation to the Clinical Stages (in absolute numbers)

	EMG Findings		
		Incorrect	
Clinical Stages	Correct	EMG Better	EMG Worse
Stage I	14	—	8
Stage II	30	2	4
Stage III	47	9	—

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option. In the case of three patients, differences were observed between the prediction of the expert system and the clinical findings as they could not fully cooperate during the examinations. One patient was deaf, and the other two were mentally disabled. This information becomes valuable in the case of the one patient for whom the prediction of the expert system and the clinical findings were completely different.

One might further argue that the limited number of 15 rules might be another source of discrepancy between the clinical results and the predicted findings of the EMG processing. However, because we are considering an expert system, these rules should cover all aspects of ankle movement as reflected by the muscular coordination. The rules classify cocontractions as disadvantageous, whereas activation confined to one muscle is interpreted as desirable in favor of an economic gait pattern. Passive movements of the ankle joint (e.g., due to gravity) are not considered in the rules. The only exception is rule 5, which refers to the dropping of the foot during the swing phase when no muscle is activated. However, the weighting of the rule 5 is low (Table 1).

The rules integrated into the expert system were based on contributions from experts in the field and from results of earlier EMG studies.[19,20,28] However, the data from the literature focused on healthy adults. This raises the question as to whether the data obtained from adults can be applied to the gait pattern of children. However, indications exist that the adult pattern of lower muscle activity may already occur by the third year of life.[16] In addition, the involved orthopedic experts have a high expertise in gait analysis of children. Another question addresses the reliability and variability of EMG data obtained from nondisabled persons. In a study by Arsenault et al.,[29] major differences with respect to the EMG outcomes were found between subjects. So far, no universal agreement has emerged on what normal gait comprises in terms of EMG signals.[28] Fuzzy logic overcomes these limitations, because it allows for some degree of uncertainty. It does not constitute distinct rules and can take differing expert views into account.

One major problem of surface EMG is the occurrence of crosstalk, which could be expected in particular between the gastrocnemius and soleus muscles. The surface EMG electrodes were located according to the SENIAM recommendations.[17] These recommendations seek to minimize the risk of crosstalk or other misinterpretation due to poor localization of the electrodes with respect to the muscle. However, even when the electrodes are optimally located, the presence of crosstalk cannot be excluded. The expert system would interpret a crosstalk signal as an EMG signal originating from the muscle to which the recording channel is attributed. However, due to the fact that fuzzy-set theory takes uncertainty into consideration, this failure will have limited impact on the results. Consequently, in applying fuzzy-set theory, the problem of crosstalk is reduced compared to other conventional ways of interpreting surface EMG signals.

Although our results show some discrepancies, we feel encouraged to compare the predictions of the expert system with the kinematic and kinetic results. Such a comparison might provide more comprehensive information. A comparison of the interpreted EMG data with the kinematic and kinetic results is certainly desirable as these results are far more accurate.

In conclusion, the interpretation of the muscular coordination pattern detected with surface electrodes can be potentially supported by an expert system based on the fuzzy-inference methodology. The example of the relatively simple ankle movement, regarding only three muscles, demonstrates clearly the feasibility of a fuzzy expert system. However, the methodology can be extended to more complex movements incorporating a higher number of muscles and joints. Initial clinical tests using the expert system show that it reliably predicts the effectiveness of ankle joint movement in gait in children with CP. Eighty percent of the predictions of the expert system corresponded to the clinical findings. We conclude that the fuzzy expert system supports the interpretation of muscular coordination patterns detected by surface EMG and makes this information accessible for introduction in the clinical routine.

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REFERENCES

1. Ferdjallah M, Wertsch JJ. 1998. Anatomical and technical consideration in surface electromyography. Phys Med Rehabil Clin N Am 9:925–931.

1554527x, 2006, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jor.20043 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2. Gage JR, DeLuca PA, Renshaw TS. 1996. Gait analysis: principles and applications with emphasis on its use in cerebral palsy. Instr Course Lect 45: 491–507.
3. Lin CJ, Guo LY, Su FC, et al. 2000. Common abnormal kinetic patterns of the knee in gait in spastic diplegia of cerebral palsy. Gait Posture 11:224–232.
4. Sutherland DH, Ohlsen R, Cooper L, et al. 1980. The development of mature gait. J Bone Joint Surg 62:336–353.
5. Damiano DL, Martellotta TL, Sullivan DJ, et al. 2000. Muscle force production and functional performance in spastic cerebral palsy. Arch Phys Med Rehabil 81:895–900.
6. Becker K, Thull B, Käsmacher-Leidinger H, et al. 1997. Design and validation of an intelligent patient monitoring and alarm system based on a fuzzy logic process model. Artif Intell Med 11:33–53.
7. Belacel N, Boulassel MR. 2001. Multicriteria fuzzy assignment method: a useful tool to assist medical diagnosis. Artif Intell Med 21:201–207.
8. Hanson CW, Marshall BE. 2001. Artificial intelligence applications in the intensive care unit. Crit Care Med 29:427–435.
9. Mahfouf M, Abbod MF, Linkens DA. 2001. A survey of fuzzy logic monitoring and control utilisation in medicine. Artif Intell Med 21:27–42.
10. Disselhorst-Klug C, Schulte E, Rau G. 2003. Assessment of the muscular co-ordination pattern using fuzzy-logic. Gait Posture 14(Suppl 2):167–169.
11. Rau G, Schulte E, Disselhorst-Klug C. 2004. From cell to movement: to what answers does EMG really contribute. J Electromyogr Kinesiol 14:611–617.
12. Maddox J. 1983. Fuzzy sets make fuzzy logic. Nature 306:637.
13. Steimann F. 1997. Fuzzy set theory in medicine. Artif Intell Med 11:1–7.
14. Zadeh LA. 2001. From computing with numbers to computing with words. Ann N Y Acad Sci 929:221–252.
15. Zadeh LA. 1965. Fuzzy sets. Inf Control 8:338–353.
16. Hermens H, Freriks B, Disselhorst-Klug C, et al. 2000. Development of recommendations for sensors and sensor placement procedures. J Electromyogr Kinesiol 10:361–374.
17. SENIAM. 1999. European recommendations for surface electromyography, results of the SENIAM project. Roessingh Research and Development B.V. Enschede, The Netherlands (www.seniam.org)
18. Mamdani EH, Assilian EH. 1975. An experiment in linguistic synthesis with a fuzzy logic controller. Int J Man–Machine Stud 7:1–13.
19. Allard P, Stokes I, Blanchi J. 1995. Three-dimensional analysis of human movement. Champaign, IL: Human Kinetics.
20. Carlsöö S. 1972. How man moves: kinesiological methods and studies. New York: Crane, Russak & Company.
21. Brunt D, Scarborough N. 1988. Ankle muscle activity during gait in children with cerebral palsy and equinovarus deformity. Arch Phys Med Rehabil 69:115–117.
22. Gage JR. 1993. Gait analysis: an essential tool in the treatment of cerebral palsy. Clin Orthop 288: 126–134.
23. Gage JR, Novacheck TF. 2001. An update on the treatment of gait problems in cerebral palsy. J Pediat Orthop-B 10:265–274.
24. Winter DA. 1984. Pathologic gait diagnosis with computer-averaged electromyographic profiles. Arch Phys Med Rehabil 65:393–398.
25. Damiano DL, Quinlivan JM, Owen BF, et al. 2002. What does the Ashworth scale really measure and are instrumented measures more valid and precise? Dev Med Child Neurol 44:112–118.
26. Pandyan AD, Johnson GR, Price CI. 1999. A review of the properties and limitations of the Ashworth and modified Ashworth Scales as measures of spasticity. Clin Rehabil 13:373–383.
27. Petersen MC, Kube DA, Palmer FB. 1998. Classification of developmental delays. Semin Pediatr Neurol 5:2–14.
28. Knutson LM, Soderberg GL. 1994. EMG: use and interpretation in gait. In: Craik RL, Oatis CA, editors. Gait analysis. Theory and application. St. Louis: Mosby.
29. Arsenault AB, Winter DA, Marteniuk RG. 1986. Is there a normal profile of EMG activity in gait? Med Biol Eng Comput 24:337–343.

1554527x, 2006, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jor.20043 by University of Dundee, Wiley Online Library on [12/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License